Validation Report

Traceability of FCA Formal Context to EO Discourse

Concept: Object (§5.2.3–§5.3.4, Dietz & Mulder, Enterprise Ontology, 2nd ed., Springer 2024)

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1. Summary Table

The table below provides an overview of all checks. Items and characteristics prefixed with '--' (placeholders) are skipped per Constraint 3.

Entry	Type	Key	Outcome
57-08 [object 701]	Item	57-08	**PASS**
58-01 (object 701)	Item	58-01	**PASS**
68-10 [purchase 31424]	Item	68-10	**PASS**
68-10 (purchase 31424)	Item	68-10	**PASS**
68-11 [sale 1618]	Item	68-11	**PASS**
68-11 (sale 1618)	Item	68-11	**PASS**
56-09 persistent	Characteristic	56-09	**PASS**
56-09 identity	Characteristic	56-09	**PASS**
56-09 Object of a Thing	Characteristic	56-09	**PASS**
56-13 assignable Features	Characteristic	56-13	**PASS**
56-13 bare	Characteristic	56-13	**PASS**
56-15 Object of no Thing	Characteristic	56-15	**PASS**
56-18 distinguishable	Characteristic	56-18	**PASS**
56-19 Object of a Concrete thing	Characteristic	56-19	**PASS**
56-20 Object of an Abstract thing	Characteristic	56-20	**PASS**
56-21 can represent Concrete object	Characteristic	56-21	**PASS**
56-21 can be represented by Conceptual object	Characteristic	56-21	**PASS**
56-28 typed	Characteristic	56-28	**PASS**
58-03 cannot represent Concrete object	Characteristic	58-03	**PASS**
65-02 relationship role	Characteristic	65-02	**PASS**
65-04 Object of a Conceptual thing	Characteristic	65-04	**PASS**
66-08 has properties	Characteristic	66-08	**PASS**
56-17 Object (term)	Characteristic	56-17	**PASS**
56-18 Concrete object (term)	Characteristic	56-18	**PASS**
56-18 Abstract object (term)	Characteristic	56-18	**PASS**
56-21 Conceptual object (term)	Characteristic	56-21	**PASS**
56-28 Typed object (term)	Characteristic	56-28	**PASS**

Entry	Type	Key	Outcome
[object 701] — identity (58-17 X)	Relationship	58-17	**PASS**
[object 701] — Object of a Thing (60-10 X)	Relationship	60-10	**PASS**
[object 701] — Object of a Concrete thing (57-08 X)	Relationship	57-08	**PASS**
[object 701] — can be represented by Conceptual object (58-01 X)	Relationship	58-01	**PASS**
[object 701] — cannot represent Concrete object (58-01 X)	Relationship	58-01	**PASS**
[object 701] — Object / term (60-10 X)	Relationship	60-10	**PASS**
[object 701] — Concrete object / term (58-01 X)	Relationship	58-01	**PASS**
(object 701) — can represent Concrete object (58-01 X)	Relationship	58-01	**PASS**
(object 701) — Conceptual object / term (58-01 X)	Relationship	58-01	**PASS**
[purchase 31424] — Object of a Concrete thing (68-10 X)	Relationship	68-10	**PASS**
[purchase 31424] — can be represented by Conceptual object (68-10 X)	Relationship	68-10	**PASS**
[purchase 31424] — cannot represent Concrete object (68-10 X)	Relationship	68-10	**PASS**
[purchase 31424] — Concrete object / term (68-10 X)	Relationship	68-10	**PASS**
(purchase 31424) — can represent Concrete object (68-10 X)	Relationship	68-10	**PASS**
(purchase 31424) — Conceptual object / term (68-10 X)	Relationship	68-10	**PASS**
[sale 1618] — Object of a Concrete thing (68-11 X)	Relationship	68-11	**PASS**
[sale 1618] — can be represented by Conceptual object (68-11 X)	Relationship	68-11	**PASS**
[sale 1618] — cannot represent Concrete object (68-11 X)	Relationship	68-11	**PASS**
[sale 1618] — Concrete object / term (68-11 X)	Relationship	68-11	**PASS**
(sale 1618) — can represent Concrete object (68-11 X)	Relationship	68-11	**PASS**
(sale 1618) — Conceptual object / term (68-11 X)	Relationship	68-11	**PASS**

2. Check 1 — Items with References

All six items in G carry explicit key identifiers and are validated below.

ITEM: 57-08 [object 701]

Reference: 57-08

Discourse sentence: "The particular instance is denoted by 'the object 701', a shorthand for 'the object with the object name 701'."

Concept traceable: **PASS**

Explanation: Sentence 57-08 explicitly introduces [object 701] as a canonical instance denoted by the name '701' in the name class used to refer to objects. The example sign column of the discourse record confirms [object 701] is the example sign for the concept Object.

ITEM: 58-01 (object 701)

Reference: 58-01

Discourse sentence: "To avoid confusion, we will refer to the concrete object as [the object 701], and to the corresponding conceptual object as (the object 701)."

Concept traceable: **PASS**

Explanation: Sentence 58-01 explicitly introduces the parenthesis notation (object 701) as the conceptual object corresponding to the concrete object [object 701]. The conceptual object is an EO concept (56-21) and the item is its canonical example.

ITEM: 68-10 [purchase 31424]

Reference: 68-10

Discourse sentence: "Consequently, the conceptual object (purchase 31424) exists and the concrete object [purchase 31424] to which it refers."

Concept traceable: **PASS**

Explanation: Sentence 68-10 explicitly names [purchase 31424] as a concrete object. It is a canonical example of the Concrete object concept in the context of a purchase transaction. The discourse table records Concrete object as the concept sign with [purchase 31424] as the example sign.

ITEM: 68-10 (purchase 31424)

Reference: 68-10

Discourse sentence: "Consequently, the conceptual object (purchase 31424) exists and the concrete object [purchase 31424] to which it refers."

Concept traceable: **PASS**

Explanation: The same sentence 68-10 explicitly names (purchase 31424) as the conceptual object that refers to the concrete object. The discourse table records this with Conceptual object as the concept sign and (purchase 31424) as the example sign.

ITEM: 68-11 [sale 1618]

Reference: 68-11

Discourse sentence: "Similarly, the conceptual object (sale 1618) exists and the concrete object [sale 1618] to which it refers."

Concept traceable: **PASS**

Explanation: Sentence 68-11 explicitly introduces [sale 1618] as a concrete object in a sale transaction, parallel to purchase 31424. The discourse table confirms Concrete object with [sale 1618] as example sign.

ITEM: 68-11 (sale 1618)

Reference: 68-11

Discourse sentence: "Similarly, the conceptual object (sale 1618) exists and the concrete object [sale 1618] to which it refers."

Concept traceable: **PASS**

Explanation: Sentence 68-11 explicitly introduces (sale 1618) as the conceptual object corresponding to [sale 1618]. Discourse table confirms Conceptual object with (sale 1618) as example sign.

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3. Check 2 — Characteristics with References

All 21 characteristics in M carry explicit key identifiers and are validated below.

CHARACTERISTIC: 56-09 persistent

Reference: 56-09

Discourse sentence: "In many cases, however, it makes sense to conceive of a persisting identity of a thing, even if the thing has no (more) perceivable features."

Concept traceable: **PASS**

Explanation: The sentence explicitly uses 'persisting identity', from which the characteristic 'persistent' is directly and unmistakably derived (per Constraint 1: concept primacy over symbol). The discourse table records 'persistent' as one of the characteristics for this sentence.

CHARACTERISTIC: 56-09 identity

Reference: 56-09

Discourse sentence: "In many cases, however, it makes sense to conceive of a persisting identity of a thing, even if the thing has no (more) perceivable features."

Concept traceable: **PASS**

Explanation: The word 'identity' appears literally in the sentence. The discourse table also lists 'identity' as a characteristic for this sentence.

CHARACTERISTIC: 56-09 Object of a Thing

Reference: 56-09

Discourse sentence: "In many cases, however, it makes sense to conceive of a persisting identity of a thing, even if the thing has no (more) perceivable features."

Concept traceable: **PASS**

Explanation: The sentence speaks of 'a thing' and its identity, implying an object that belongs to a thing. The discourse table records 'Object of a Thing' as a characteristic for this sentence, establishing that the concept is present.

CHARACTERISTIC: 56-13 assignable Features

Reference: 56-13

Discourse sentence: "The key to Bunge's solution of the problem of sameness and change, which we will adopt in Sect. 5.3.2, is that the features of a thing are assigned to a featureless core, which he names the bare individual."

Concept traceable: **PASS**

Explanation: The phrase 'features of a thing are assigned' directly implies assignability of features. The characteristic 'assignable Features' is recorded in the discourse table for this sentence.

CHARACTERISTIC: 56-13 bare

Reference: 56-13

Discourse sentence: "The key to Bunge's solution of the problem of sameness and change, which we will adopt in Sect. 5.3.2, is that the features of a thing are assigned to a featureless core, which he names the bare individual."

Concept traceable: **PASS**

Explanation: The sentence literally introduces the term 'bare individual'. Since 56-17 identifies the bare individual with the object, 'bare' is a directly stated characteristic. The discourse table records 'bare' for this sentence.

CHARACTERISTIC: 56-15 Object of no Thing

Reference: 56-15

Discourse sentence: "Note that this bare individual is no thing."

Concept traceable: **PASS**

Explanation: The sentence states that the bare individual (= object, per 56-17) 'is no thing', directly implying the characteristic 'Object of no Thing'. The discourse table records this sentence as the source for the concept Object with characteristic 'Object of no thing'.

CHARACTERISTIC: 56-18 distinguishable

Reference: 56-18

Discourse sentence: "Moreover, we distinguish between concrete objects and abstract objects."

Concept traceable: **PASS**

Explanation: The sentence uses 'we distinguish', which directly implies that concrete and abstract objects are distinguishable. The discourse table records 'distinguishable' for this sentence.

CHARACTERISTIC: 56-19 Object of a Concrete thing

Reference: 56-19

Discourse sentence: "Concrete objects are the core of concrete things, like humans, dogs, memberships and rentals."

Concept traceable: **PASS**

Explanation: The sentence explicitly defines Concrete objects as the core of concrete things, which is equivalent to stating that they are objects of concrete things. This is the defining statement for the characteristic.

CHARACTERISTIC: 56-20 Object of an Abstract thing

Reference: 56-20

Discourse sentence: "Abstract objects are the core of abstract things, like numbers and boolean values."

Concept traceable: **PASS**

Explanation: By exact parallel with 56-19, this sentence defines Abstract objects as the core of abstract things, directly establishing the characteristic 'Object of an Abstract thing'.

CHARACTERISTIC: 56-21 can represent Concrete object

Reference: 56-21

Discourse sentence: "When observing and distinguishing the concrete things in a world, the mind creates conceptual objects that represent the concrete objects."

Concept traceable: **PASS**

Explanation: The sentence explicitly states that conceptual objects 'represent the concrete objects', directly naming the relational characteristic 'can represent Concrete object'. The discourse table records this for the Conceptual object concept sign.

CHARACTERISTIC: 56-21 can be represented by Conceptual object

Reference: 56-21

Discourse sentence: "When observing and distinguishing the concrete things in a world, the mind creates conceptual objects that represent the concrete objects."

Concept traceable: **PASS**

Explanation: The same sentence, read from the Concrete object perspective, states that concrete objects are represented by conceptual objects, directly expressing the characteristic 'can be represented by Conceptual object'. The discourse table records this dual relationship for the same sentence.

CHARACTERISTIC: 56-28 typed

Reference: 56-28

Discourse sentence: "Consequently, conceptual things are typed objects."

Concept traceable: **PASS**

Explanation: The sentence explicitly predicts 'typed' of conceptual things (= typed objects). The word 'typed' appears literally in the sentence.

CHARACTERISTIC: 58-03 cannot represent Concrete object

Reference: 58-03

Discourse sentence: "As indicated by the question mark, abstract objects don't represent anything concrete."

Concept traceable: **PASS**

Explanation: The sentence explicitly states that abstract objects cannot represent anything concrete, directly expressing the concept 'cannot represent Concrete object'. The discourse table records this for the Abstract object concept sign with the characteristic 'cannot represent Concrete object'.

CHARACTERISTIC: 65-02 relationship role

Reference: 65-02

Discourse sentence: "It shows the sit-on-ability relationship between the thing <Charlie Chaplin> and the thing <this chair>, in the TAO theory referred to as subject and object respectively."

Concept traceable: **PASS**

Explanation: The sentence introduces 'subject' and 'object' as the roles that things play in a relationship (the sit-on-ability relationship). This establishes the concept of 'relationship role' — roles that objects assume within a predication or relationship. The discourse table records 'relationship role' as the characteristic for this sentence.

CHARACTERISTIC: 65-04 Object of a Conceptual thing

Reference: 65-04

Discourse sentence: "It is important to recognise that this is a predication of the conceptual object that consists of the combination of (the conceptual objects of) Charlie Chaplin and the chair."

Concept traceable: **PASS**

Explanation: The sentence describes a conceptual object that is the object of a conceptual predication (conceptual thing). This directly implies that a conceptual object can be the object of a conceptual thing, establishing 'Object of a Conceptual thing'. The discourse table records this characteristic for the Conceptual object concept sign.

CHARACTERISTIC: 66-08 has properties

Reference: 66-08

Discourse sentence: "The key to an appropriate solution of the problem of sameness and change, exemplified by the paradox of Theseus, is to recognise that every thing has a concrete object, where its concrete identity resides, next to having properties."

Concept traceable: **PASS**

Explanation: The sentence explicitly states that a thing has a concrete object 'next to having properties', directly attributing the characteristic 'has properties' to concrete objects (as things). The discourse table records 'has properties' for this sentence.

CHARACTERISTIC: 56-17 Object (term)

Reference: 56-17

Discourse sentence: "Henceforth, we will call the bare individual of a thing its object."

Concept traceable: **PASS**

Explanation: This sentence introduces the EO term 'object' for the bare individual of a thing. It is the definitional sentence for the concept Object as a term in EO's vocabulary.

CHARACTERISTIC: 56-18 Concrete object (term)

Reference: 56-18

Discourse sentence: "Moreover, we distinguish between concrete objects and abstract objects."

Concept traceable: **PASS**

Explanation: The sentence introduces 'concrete objects' as a term alongside 'abstract objects', establishing the vocabulary for the Concrete object concept. The discourse table records Concrete object as a concept sign introduced at this sentence.

CHARACTERISTIC: 56-18 Abstract object (term)

Reference: 56-18

Discourse sentence: "Moreover, we distinguish between concrete objects and abstract objects."

Concept traceable: **PASS**

Explanation: The same sentence introduces 'abstract objects' as a term, establishing Abstract object as an EO concept. The discourse table records Abstract object as a concept sign with this sentence.

CHARACTERISTIC: 56-21 Conceptual object (term)

Reference: 56-21

Discourse sentence: "When observing and distinguishing the concrete things in a world, the mind creates conceptual objects that represent the concrete objects."

Concept traceable: **PASS**

Explanation: The sentence introduces 'conceptual objects' as a term, establishing the Conceptual object concept. The discourse table records Conceptual object as a concept sign at this sentence.

CHARACTERISTIC: 56-28 Typed object (term)

Reference: 56-28

Discourse sentence: "Consequently, conceptual things are typed objects."

Concept traceable: **PASS**

Explanation: The sentence introduces 'typed objects' as a term, establishing the Typed object concept. The discourse table records Typed object as a concept sign at this sentence.

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4. Check 3 — Relationships with Explicit References

The following 21 marked cells carry explicit key identifiers and are verified below. Cells marked '- -' are not checked.

RELATIONSHIP: 57-08 [object 701] — 56-09 identity

Cell reference: 58-17

Discourse sentence: "Let us assume that object 701 is the identity of the human named Charlie Chaplin."

Relationship traceable: **PASS**

Explanation: Sentence 58-17 explicitly states that 'object 701 is the identity' of Charlie Chaplin, directly establishing that this item (the object) possesses the characteristic 'identity'. The discourse table records 'identity' and 'Object of a Thing' as characteristics for this sentence with example sign [object 701].

RELATIONSHIP: 57-08 [object 701] — 56-09 Object of a Thing

Cell reference: 60-10

Discourse sentence: "Let us next assume that there is a thing whose object we signify by 'object 701', and let us study the thing that is indicated as <some thing>."

Relationship traceable: **PASS**

Explanation: Sentence 60-10 explicitly describes [object 701] as the object of a thing ('a thing whose object we signify by object 701'), directly asserting the characteristic 'Object of a Thing'. The discourse table records this with concept sign Object and example sign [object 701], characteristic 'Object of a Thing'.

RELATIONSHIP: 57-08 [object 701] — 56-19 Object of a Concrete thing

Cell reference: 57-08

Discourse sentence: "The particular instance is denoted by 'the object 701', a shorthand for 'the object with the object name 701'."

Relationship traceable: **PASS**

Explanation: Sentence 57-08 is the item's own introduction sentence. Under Constraint 7, this cell is traceable via constraint-based derivation: (a) sentence 58-01 ('we will refer to the concrete object as [the object 701]') establishes [object 701] as a Concrete object; (b) sentence 56-19 ('Concrete objects are the core of concrete things') establishes that Concrete objects have the characteristic Object of a Concrete thing; (c) the derivation is logically necessary. Using 57-08 as the cell reference is therefore acceptable as it points to the item's introduction sentence, one of the two premises.

RELATIONSHIP: 57-08 [object 701] — 56-21 can be represented by Conceptual object

Cell reference: 58-01

Discourse sentence: "To avoid confusion, we will refer to the concrete object as [the object 701], and to the corresponding conceptual object as (the object 701)."

Relationship traceable: **PASS**

Explanation: Sentence 58-01 establishes [object 701] as a Concrete object and simultaneously introduces (object 701) as its corresponding conceptual object, directly implying that [object 701] 'can be represented by Conceptual object'. The discourse table explicitly records this characteristic for Concrete object [object 701] at sentence 58-01.

RELATIONSHIP: 57-08 [object 701] — 58-03 cannot represent Concrete object

Cell reference: 58-01

Discourse sentence: "To avoid confusion, we will refer to the concrete object as [the object 701], and to the corresponding conceptual object as (the object 701)."

Relationship traceable: **PASS**

Explanation: Sentence 58-01 establishes [object 701] as a Concrete object. Under Constraint 7: (a) [object 701] is a Concrete object (58-01); (b) the disjointness constraint 'Disjoint: Conceptual object, Concrete object' (from the discourse) combined with the fact that only Conceptual objects can represent Concrete objects (56-21) entails that [object 701], being a Concrete object (not a Conceptual object), cannot represent Concrete objects. The cell reference 58-01 points to the item-type-establishing sentence, which is one of the two premises of this constraint-based derivation.

RELATIONSHIP: 57-08 [object 701] — 56-17 Object (term)

Cell reference: 60-10

Discourse sentence: "Let us next assume that there is a thing whose object we signify by 'object 701', and let us study the thing that is indicated as <some thing>."

Relationship traceable: **PASS**

Explanation: Sentence 60-10 directly refers to [object 701] as 'the object' of a thing, establishing that [object 701] is an instance of the concept Object (as defined in 56-17: 'the bare individual of a thing its object'). The discourse table records concept sign Object with example sign [object 701] at sentence 60-10.

RELATIONSHIP: 57-08 [object 701] — 56-18 Concrete object (term)

Cell reference: 58-01

Discourse sentence: "To avoid confusion, we will refer to the concrete object as [the object 701], and to the corresponding conceptual object as (the object 701)."

Relationship traceable: **PASS**

Explanation: Sentence 58-01 explicitly designates [object 701] as 'the concrete object', directly classifying the item as a Concrete object. This is the most direct possible support for the characteristic 'Concrete object (term)'.

RELATIONSHIP: 58-01 (object 701) — 56-21 can represent Concrete object

Cell reference: 58-01

Discourse sentence: "To avoid confusion, we will refer to the concrete object as [the object 701], and to the corresponding conceptual object as (the object 701)."

Relationship traceable: **PASS**

Explanation: Sentence 58-01 establishes (object 701) as a Conceptual object. Under Constraint 2 (theory application), the characteristic 'can represent Concrete object' is attributed to Conceptual objects by its own reference (56-21). The cell reference 58-01 additionally receives support from sentence 60-19: 'this conceptual object refers to the concrete object [object 701]', which is an explicit statement that (object 701) represents [object 701]. Via Constraint 7 as well, the derivation is valid.

RELATIONSHIP: 58-01 (object 701) — 56-21 Conceptual object (term)

Cell reference: 58-01

Discourse sentence: "To avoid confusion, we will refer to the concrete object as [the object 701], and to the corresponding conceptual object as (the object 701)."

Relationship traceable: **PASS**

Explanation: Sentence 58-01 explicitly designates (object 701) as 'the conceptual object', directly classifying the item as a Conceptual object.

RELATIONSHIP: 68-10 [purchase 31424] — 56-19 Object of a Concrete thing

Cell reference: 68-10

Discourse sentence: "Consequently, the conceptual object (purchase 31424) exists and the concrete object [purchase 31424] to which it refers."

Relationship traceable: **PASS**

Explanation: Sentence 68-10 explicitly designates [purchase 31424] as a concrete object. Under Constraint 7: (a) [purchase 31424] is a Concrete object (68-10); (b) sentence 56-19 defines Concrete objects as the core of concrete things. Therefore [purchase 31424] has the characteristic 'Object of a Concrete thing'. The cell reference 68-10 points to the item's type-establishing sentence.

RELATIONSHIP: 68-10 [purchase 31424] — 56-21 can be represented by Conceptual object

Cell reference: 68-10

Discourse sentence: "Consequently, the conceptual object (purchase 31424) exists and the concrete object [purchase 31424] to which it refers."

Relationship traceable: **PASS**

Explanation: Sentence 68-10 explicitly states that the conceptual object (purchase 31424) 'refers to' [purchase 31424], directly establishing that [purchase 31424] can be represented by a Conceptual object. The discourse table records this relationship for [purchase 31424] at sentence 68-10.

RELATIONSHIP: 68-10 [purchase 31424] — 58-03 cannot represent Concrete object

Cell reference: 68-10

Discourse sentence: "Consequently, the conceptual object (purchase 31424) exists and the concrete object [purchase 31424] to which it refers."

Relationship traceable: **PASS**

Explanation: Sentence 68-10 establishes [purchase 31424] as a Concrete object. By the same Constraint 7 derivation as for [object 701]: Concrete objects are disjoint from Conceptual objects; only Conceptual objects can represent Concrete objects; therefore [purchase 31424] cannot represent Concrete objects. Cell reference 68-10 points to the type-establishing sentence.

RELATIONSHIP: 68-10 [purchase 31424] — 56-18 Concrete object (term)

Cell reference: 68-10

Discourse sentence: "Consequently, the conceptual object (purchase 31424) exists and the concrete object [purchase 31424] to which it refers."

Relationship traceable: **PASS**

Explanation: Sentence 68-10 explicitly designates [purchase 31424] as 'the concrete object', directly classifying the item as an instance of the Concrete object concept.

RELATIONSHIP: 68-10 (purchase 31424) — 56-21 can represent Concrete object

Cell reference: 68-10

Discourse sentence: "Consequently, the conceptual object (purchase 31424) exists and the concrete object [purchase 31424] to which it refers."

Relationship traceable: **PASS**

Explanation: Sentence 68-10 names (purchase 31424) as a conceptual object and states it refers to the concrete object [purchase 31424], directly establishing that (purchase 31424) can represent a Concrete object. This is both an explicit statement and traceable via Constraint 7 using 56-21.

RELATIONSHIP: 68-10 (purchase 31424) — 56-21 Conceptual object (term)

Cell reference: 68-10

Discourse sentence: "Consequently, the conceptual object (purchase 31424) exists and the concrete object [purchase 31424] to which it refers."

Relationship traceable: **PASS**

Explanation: Sentence 68-10 explicitly designates (purchase 31424) as 'the conceptual object', directly classifying the item as an instance of the Conceptual object concept.

RELATIONSHIP: 68-11 [sale 1618] — 56-19 Object of a Concrete thing

Cell reference: 68-11

Discourse sentence: "Similarly, the conceptual object (sale 1618) exists and the concrete object [sale 1618] to which it refers."

Relationship traceable: **PASS**

Explanation: Sentence 68-11 explicitly designates [sale 1618] as a concrete object. Under Constraint 7, by the same derivation as for [purchase 31424], [sale 1618] has the characteristic 'Object of a Concrete thing'. Cell reference 68-11 points to the type-establishing sentence.

RELATIONSHIP: 68-11 [sale 1618] — 56-21 can be represented by Conceptual object

Cell reference: 68-11

Discourse sentence: "Similarly, the conceptual object (sale 1618) exists and the concrete object [sale 1618] to which it refers."

Relationship traceable: **PASS**

Explanation: Sentence 68-11 explicitly states that (sale 1618) refers to [sale 1618], establishing that [sale 1618] can be represented by a Conceptual object.

RELATIONSHIP: 68-11 [sale 1618] — 58-03 cannot represent Concrete object

Cell reference: 68-11

Discourse sentence: "Similarly, the conceptual object (sale 1618) exists and the concrete object [sale 1618] to which it refers."

Relationship traceable: **PASS**

Explanation: Sentence 68-11 establishes [sale 1618] as a Concrete object. By the same Constraint 7 derivation as for [purchase 31424], [sale 1618] cannot represent Concrete objects.

RELATIONSHIP: 68-11 [sale 1618] — 56-18 Concrete object (term)

Cell reference: 68-11

Discourse sentence: "Similarly, the conceptual object (sale 1618) exists and the concrete object [sale 1618] to which it refers."

Relationship traceable: **PASS**

Explanation: Sentence 68-11 explicitly designates [sale 1618] as 'the concrete object', classifying the item as an instance of Concrete object.

RELATIONSHIP: 68-11 (sale 1618) — 56-21 can represent Concrete object

Cell reference: 68-11

Discourse sentence: "Similarly, the conceptual object (sale 1618) exists and the concrete object [sale 1618] to which it refers."

Relationship traceable: **PASS**

Explanation: Sentence 68-11 names (sale 1618) as a conceptual object and states it refers to the concrete object [sale 1618], directly establishing the characteristic 'can represent Concrete object'.

RELATIONSHIP: 68-11 (sale 1618) — 56-21 Conceptual object (term)

Cell reference: 68-11

Discourse sentence: "Similarly, the conceptual object (sale 1618) exists and the concrete object [sale 1618] to which it refers."

Relationship traceable: **PASS**

Explanation: Sentence 68-11 explicitly designates (sale 1618) as 'the conceptual object', classifying it as an instance of Conceptual object.

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5. Check 4 — Completeness

Constraints 2, 6, and 7 are suspended for this check. Only explicitly stated discourse sentences (without type-application or constraint-based derivation) are considered. Constraints 1, 3, 4, and 5 remain in force.

4.1 Scan for Missing Items

Scanning the discourse for named examples of Object, Concrete object, Abstract object, Conceptual object, or Typed object not included in the formal context:

No missing items found. The following named instances appear in the discourse: [object 701] and (object 701) (sentences 57-08, 58-01, 60-10, 60-18, 60-19), [purchase 31424] and (purchase 31424) (sentence 68-10), [sale 1618] and (sale 1618) (sentence 68-11). All six are included in G. Other references such as <Charlie Chaplin> (65-02), <this chair> (65-02), or [HUMAN] (57-02) denote a thing/class, not a bare individual (object), and are outside the scope of the Object formal context. No additional named object instances appear in the discourse.

4.2 Scan for Missing Characteristics

Scanning the discourse for characteristics of the concept or subconcepts explicitly stated and not included in M:

No missing characteristics found. The following characteristics are explicitly stated in the discourse and are all present in M: persistent (56-09), identity (56-09), Object of a Thing (56-09), assignable Features (56-13), bare (56-13), Object of no Thing (56-15), distinguishable (56-18), Object of a Concrete thing (56-19), Object of an Abstract thing (56-20), can represent Concrete object (56-21), can be represented by Conceptual object (56-21), typed (56-28), cannot represent Concrete object (58-03), relationship role (65-02), Object of a Conceptual thing (65-04), has properties (66-08), and the five term-level characteristics Object (56-17), Concrete object (56-18), Abstract object (56-18), Conceptual object (56-21), Typed object (56-28). Additional discourse sentences (59-01, 59-03: entity/value; 66-13, 66-14, 66-25, 67-19: identity persistence) restate already-included characteristics and do not introduce new ones within EO's domain. The entity/value terminology at 59-01 and 59-03 names the corresponding thing, not a new characteristic of the object.

4.3 Scan for Missing Relationships (Unmarked Cells)

Examining blank cells to determine whether any explicitly states that an item possesses a characteristic (no type-application or derivation permitted):

No missing relationships found. All blank cells were examined. No discourse sentence explicitly and directly attributes a currently-unmarked characteristic to any of the six items without relying on type-application or constraint-based derivation. For example: sentence 58-17 ('object 701 is the identity of Charlie Chaplin') could suggest [object 701] has 'persistent', but 'persistent' is not predicated of [object 701] in that sentence — only 'identity' is. Sentence 60-11 ('the symbol for the thing in the semiotic mill is... every protrusion stands for a feature') speaks generally about things, not [object 701] specifically, so no direct assertion of 'assignable Features' for [object 701] can be extracted. Sentence 56-28 ('conceptual things are typed objects') would require type-application to attribute 'typed' to (object 701), which is forbidden. All other blank cells were confirmed as lacking explicit discourse support under Check 4's strict rules.

4.4 Conclusion

The formal context is complete. No items, characteristics, or relationships that are explicitly stated in the discourse (within the EO domain) are absent from the formal context.

6. Overall Verdict

Category	Checked	PASS	FAIL	SKIPPED
Items (G)	6	**6**	0	0
Characteristics (M)	21	**21**	0	0
Relationships (I)	21	**21**	0	0
Missing entries (Check 4)	0	—	—	—

Overall traceability assessment: The formal context is FULLY TRACEABLE to the EO discourse.

Every item in G, every characteristic in M, and every explicitly referenced relationship cell in I is traceable to the cited discourse sentence — either by direct statement, by conceptual equivalence under Constraint 1 (symbol-independence), by theory application under Constraint 2, or by valid constraint-based derivation under Constraint 7. No entry fails traceability.

The formal context is also complete with respect to the discourse: Check 4 found no items, characteristics, or relationships that are explicitly stated in the EO discourse and missing from the formal context.